

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

500 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

500 KWP GRID-TIED SOLAR PLANT

Net investment ₹1.93 Cr after input tax credit, generating 7,66,500 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.7 yrs

Post-tax, undiscounted

PROJECT IRR

27.5%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.15

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹3.56 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	500 kWp
Year 1 generation	7,66,500 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹54.88 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹40.50 / Wp (incl. GST)
Project price incl. GST	₹2.02 Cr
GST at 5% — recovered as ITC	₹9.64 L
Net capitalised cost	₹1.93 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹48.54 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,92,85,715	77,14,286	1,15,71,429	19,41,532
2	1,15,71,428	46,28,571	69,42,857	11,64,919
3	69,42,857	27,77,143	41,65,714	6,98,951
4	41,65,715	16,66,286	24,99,429	4,19,371
5	24,99,428	9,99,771	14,99,657	2,51,622
6	14,99,657	5,99,863	8,99,794	1,50,974
7	8,99,795	3,59,918	5,39,877	90,584
8	5,39,877	2,15,951	3,23,926	54,351
9	3,23,926	1,29,570	1,94,356	32,610
10	1,94,355	77,742	1,16,613	19,566
11	1,16,613	46,645	69,968	11,740
12	69,968	27,987	41,981	7,044
Tax shield, years 1-12				48,43,263

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30-year post-tax cash flow

500 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	7,66,500	54,88,140	—	77,14,286	(5,60,276)	60,48,416	60,48,416
2	7,62,668	56,79,127	3,00,000	46,28,571	1,88,900	51,90,227	1,12,38,643
3	7,58,854	58,76,761	3,00,000	27,77,143	7,04,608	48,72,153	1,61,10,796
4	7,55,060	60,81,272	3,00,000	16,66,286	10,35,660	47,45,612	2,08,56,408
5	7,51,285	62,92,900	3,00,000	9,99,771	12,56,671	47,36,230	2,55,92,638
6	7,47,528	65,11,893	3,00,000	5,99,863	14,12,436	47,99,458	3,03,92,096
7	7,43,791	67,38,507	3,00,000	3,59,918	15,29,859	49,08,648	3,53,00,744
8	7,40,072	69,73,007	3,00,000	2,15,951	16,25,112	50,47,895	4,03,48,639
9	7,36,371	72,15,668	3,00,000	1,29,570	17,07,925	52,07,743	4,55,56,382
10	7,32,689	74,66,773	3,00,000	77,742	17,84,167	53,82,606	5,09,38,988
11	7,29,026	77,26,617	3,00,000	46,645	18,57,391	55,69,226	5,65,08,214
12	7,25,381	79,95,503	3,00,000	27,987	19,29,760	57,65,743	6,22,73,957
13	7,21,754	82,73,747	3,00,000	16,792	20,02,606	59,71,140	6,82,45,097
14	7,18,145	85,61,673	3,00,000	10,075	20,76,762	61,84,911	7,44,30,008
15	7,14,554	88,59,619	3,00,000	6,045	21,52,764	64,06,856	8,08,36,864
16	7,10,982	91,67,934	3,00,000	3,627	22,30,969	66,36,965	8,74,73,829
17	7,07,427	94,86,978	3,00,000	2,176	23,11,631	68,75,347	9,43,49,176
18	7,03,890	98,17,125	3,00,000	1,306	23,94,941	71,22,184	10,14,71,360
19	7,00,370	1,01,58,761	3,00,000	783	24,81,056	73,77,705	10,88,49,065
20	6,96,868	1,05,12,286	3,00,000	470	25,70,110	76,42,176	11,64,91,241
21	6,93,384	1,08,78,113	3,00,000	282	26,62,229	79,15,885	12,44,07,126
22	6,89,917	1,12,56,672	3,00,000	169	27,57,533	81,99,139	13,26,06,265
23	6,86,467	1,16,48,404	3,00,000	102	28,56,141	84,92,263	14,10,98,528
24	6,83,035	1,20,53,768	3,00,000	61	29,58,173	87,95,595	14,98,94,123
25	6,79,620	1,24,73,240	3,00,000	37	30,63,752	91,09,488	15,90,03,611
26	6,76,222	1,29,07,308	3,00,000	22	31,73,002	94,34,306	16,84,37,917
27	6,72,841	1,33,56,483	3,00,000	13	32,86,052	97,70,430	17,82,08,347
28	6,69,477	1,38,21,288	3,00,000	8	34,03,036	1,01,18,252	18,83,26,599
29	6,66,129	1,43,02,269	3,00,000	5	35,24,090	1,04,78,179	19,88,04,778
30	6,62,798	1,47,99,988	3,00,000	3	36,49,356	1,08,50,632	20,96,55,410
Total	2,14,03,105	28,23,81,824	87,00,000	1,92,85,709	6,40,26,416	20,96,55,410	20,96,55,410

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

500 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹3.56 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹20.97 Cr
30-year generation	2,14,03,103 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,14,03,103 units, avoiding 17,550.5 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.02 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 500 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.